

Result: 58% 7 / 12	 101810 001178	Neptun ID: Name: _____	Draft papers: ☐
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The test consists of **4 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

1.) (essay)

2.) (essay)

3.)

a	b	c	d
☐	☐	☐	☐

4.)

☐	☐	☐	☐
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Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Define the semantic value of $(A \equiv B)$!

Score: 2 / 2.

2.) Define the elementary disjunction.

Score: 0 / 2.

3.) Which statements are true for the following three formula?

$$A \Rightarrow q \wedge (q \wedge q), \quad B \Rightarrow \neg q \supset \neg p, \quad C \Rightarrow q \vee p$$

-1 $A \Leftrightarrow B,$

2 $\{A, B, C\}$ is satisfiable,

0 $A, B \models C,$

(d) $(A \vee B)$ is valid.

4.) Which statements are true for the following three formula?

$$A \Rightarrow q \wedge p, \quad B \Rightarrow q \vee \neg(\neg p \supset q), \quad C \Rightarrow q \equiv q$$

(a) $A \Leftrightarrow B,$

2 $\{A, B, C\}$ is satisfiable,

2 $A, B \models C,$

(d) $(A \vee B)$ is valid.